Web Application Penetration Testing (OWASP Top 10 Focus)

Week-3 & Week-4 Project Report:-

Week 3 – Advanced Attacks

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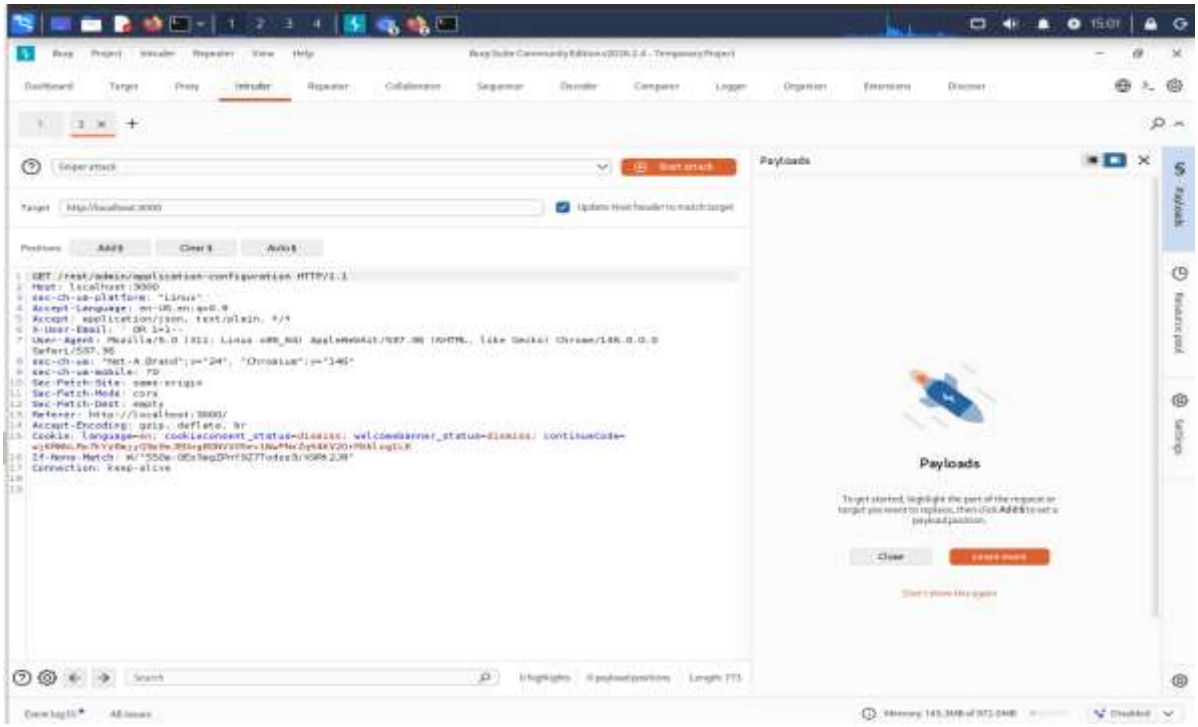
1. Broken Access Control

Steps

- Intercepted request in Burp
- Modified header:

X-User-Email: ' OR 1=1--

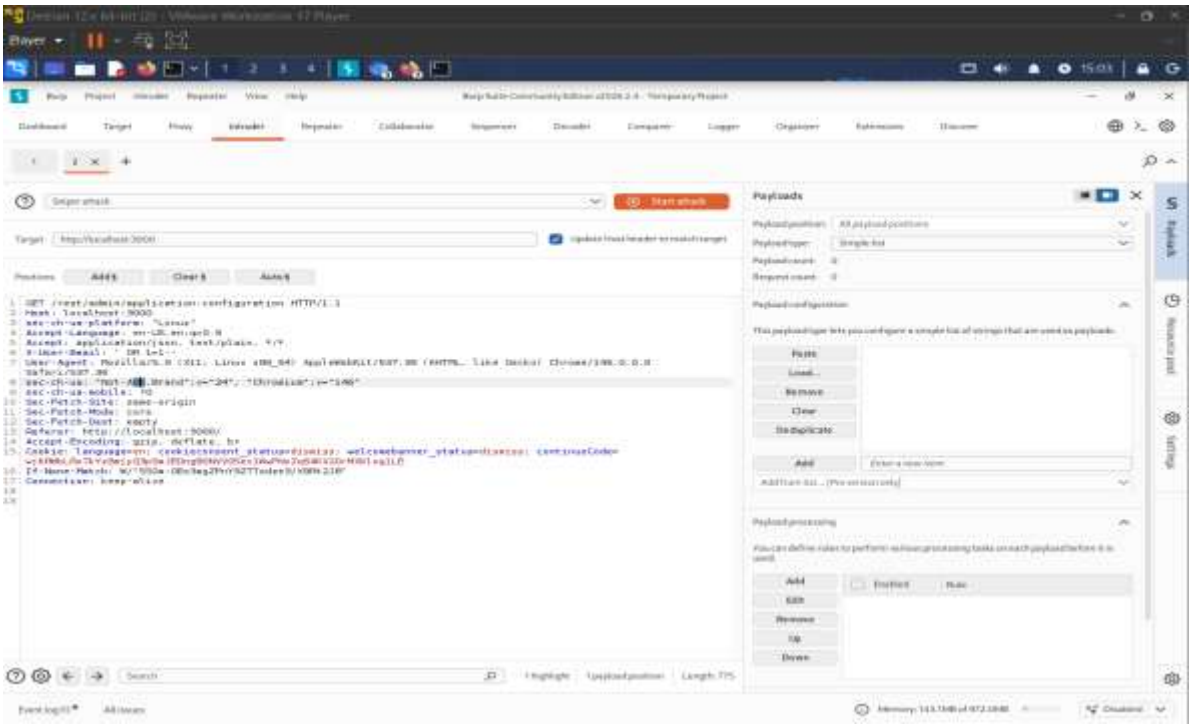
- Modified request in Burp Intruder



Result

- Accessed admin configuration endpoint

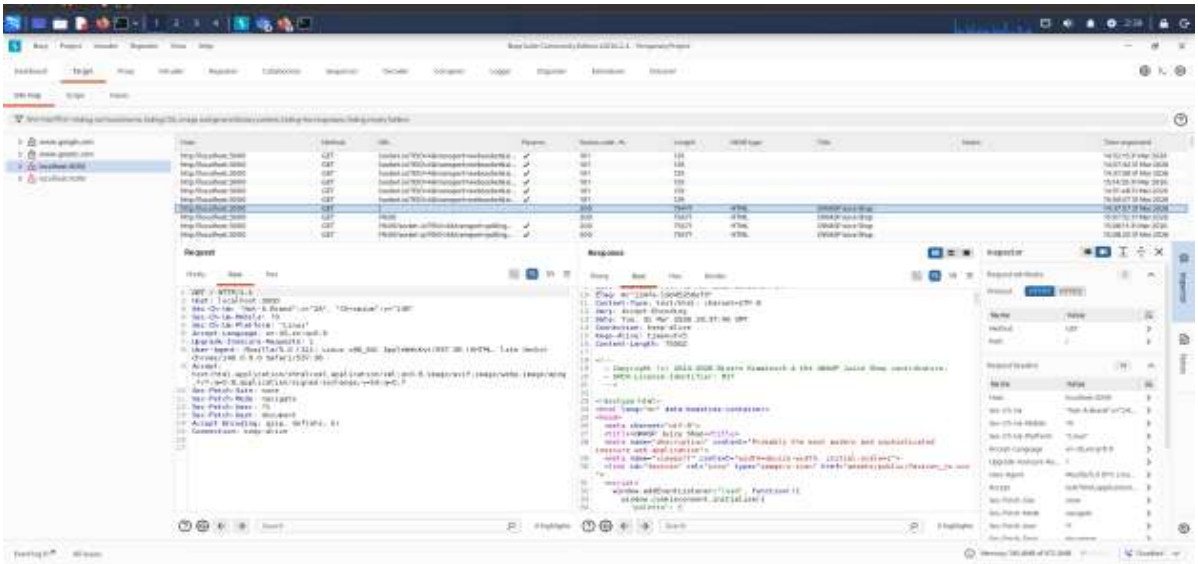
- Admin API response (unauthorized access)



2. Security Misconfiguration

Findings

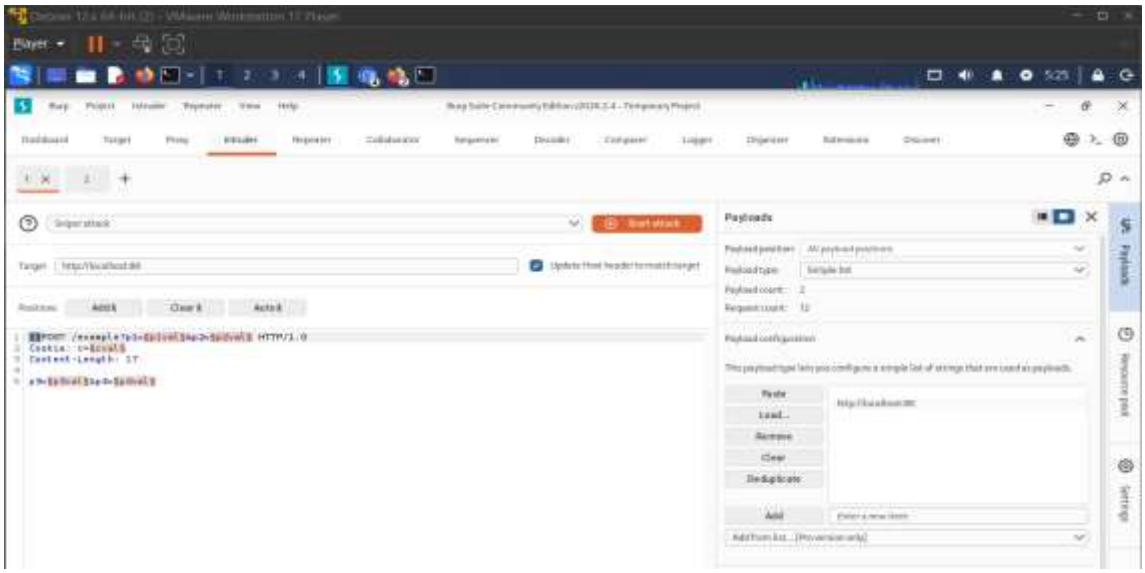
- Verbose HTTP responses
- Internal application data exposed



3. Burp Intruder Testing

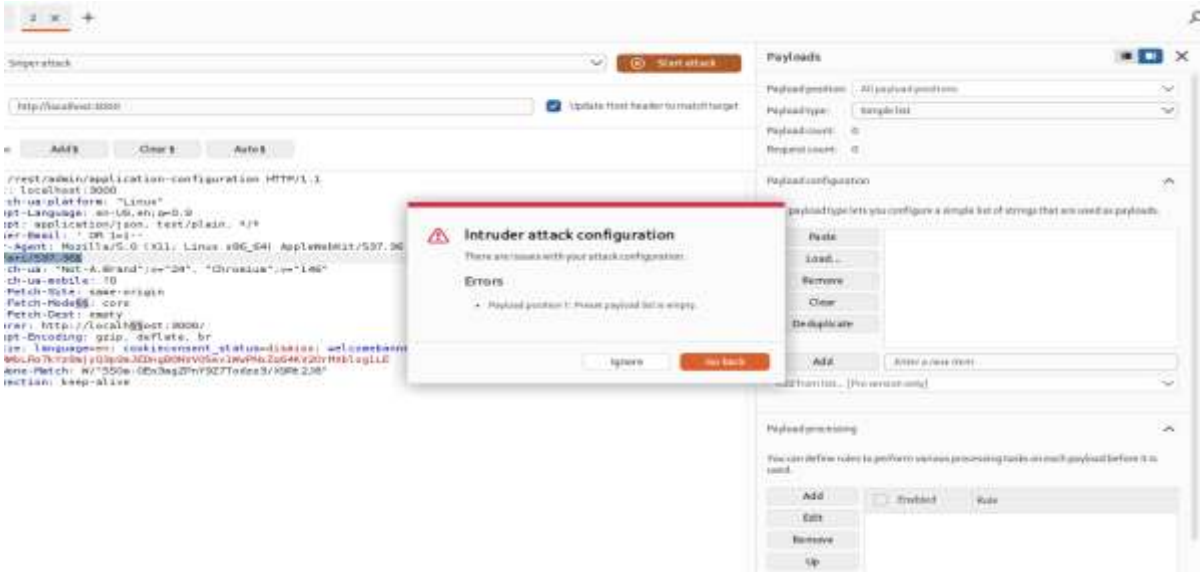
Steps

- Sent request to Intruder
- Configured payload positions
- Intruder attack setup



Issue Observed

- Payload list initially empty
- Intruder configuration error



PROOF OF CONCEPT (PoC)

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01 - Authentication Bypass via SQL Injection

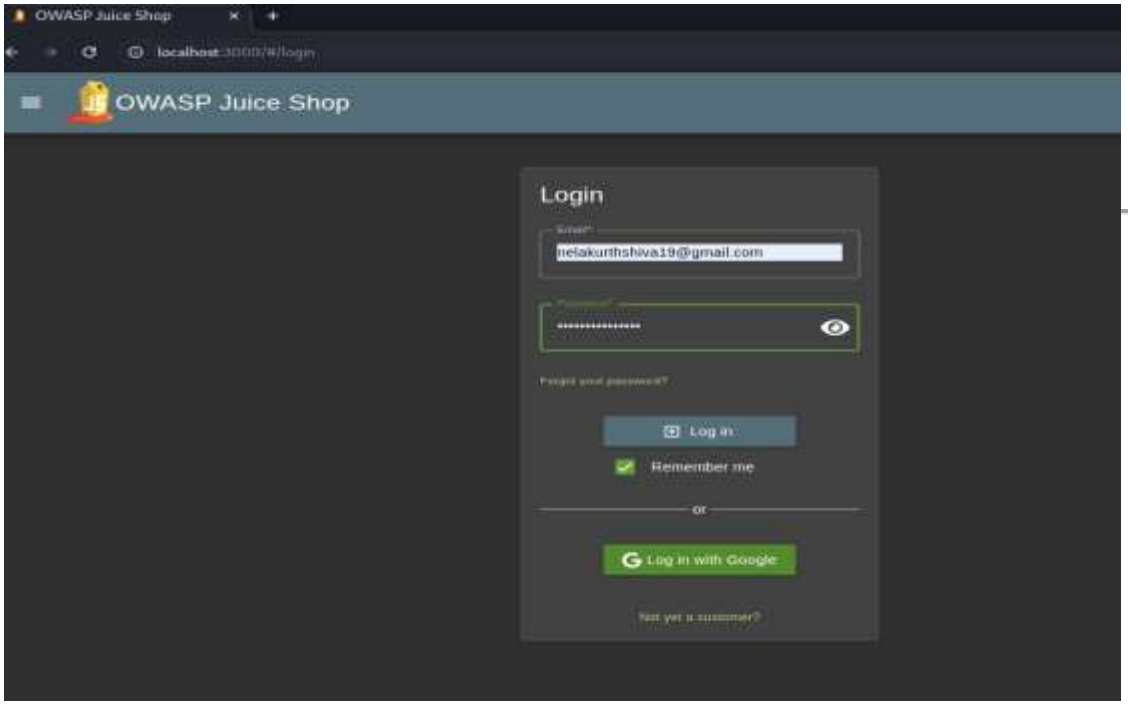
Description:
The login functionality of the application is vulnerable to SQL Injection due to improper input validation. The application directly processes user-supplied input in the email field without sanitization, allowing manipulation of the backend SQL query.

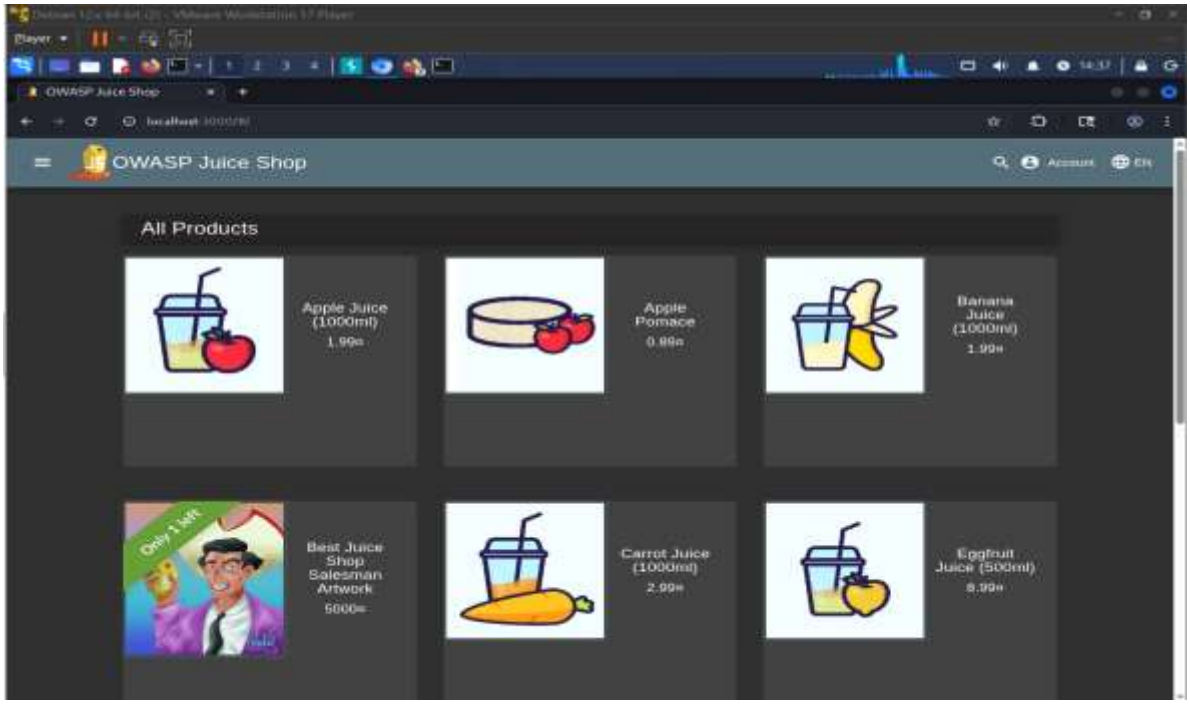
Impact (High Risk):
An attacker can bypass authentication and gain administrative access without valid credentials, leading to full system compromise.

Steps to Reproduce:

1. Navigate to the login page: http://localhost:3000/#/login
2. Enter the following payload in the Email field:
   ' OR 1=1 --
3. Enter any random password (e.g., test)
4. Click on **Log In**
5. Observe successful login without valid credentials

Screenshot Explanation:





02 - Cross-Site Scripting (XSS)

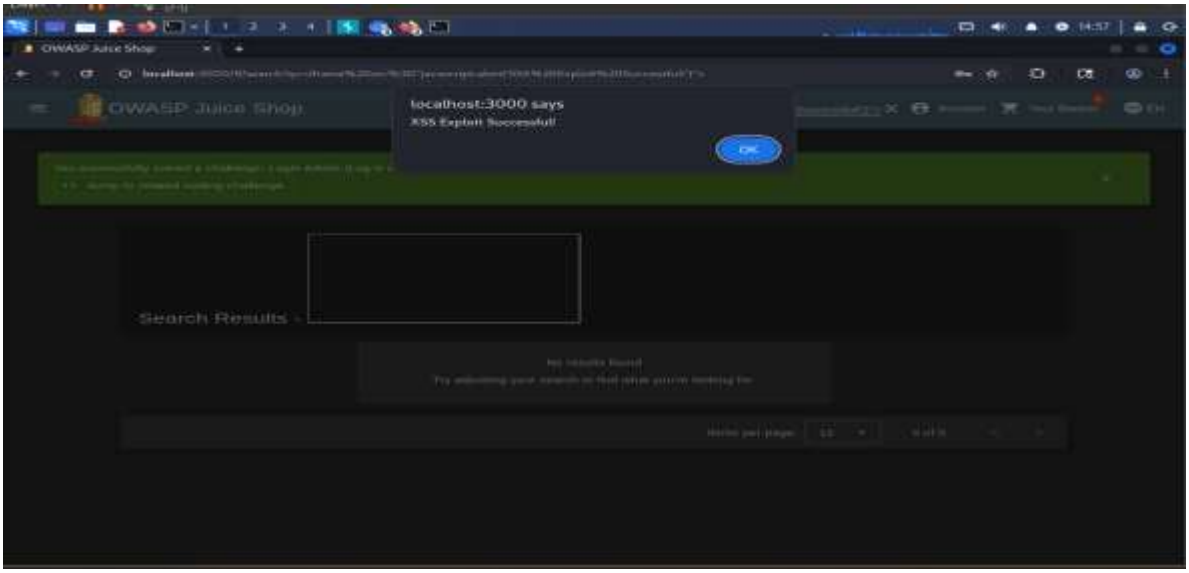
Description:
The application is vulnerable to Cross-Site Scripting (XSS), where user input is not properly sanitized before being rendered in the browser. This allows execution of malicious JavaScript code.

Impact (High Risk):
Attackers can execute scripts in the victim's browser, leading to session hijacking, data theft, or phishing attacks.

Steps to Reproduce:

1. Navigate to the search bar in the application
2. Enter the following payload:
   <iframe src="javascript:alert('XSS Exploit Successful!')">
3. Submit the search request
4. Observe that a JavaScript alert is executed

Screenshot Explanation:



03 - Sensitive Data Exposure (Exposed /ftp Directory)

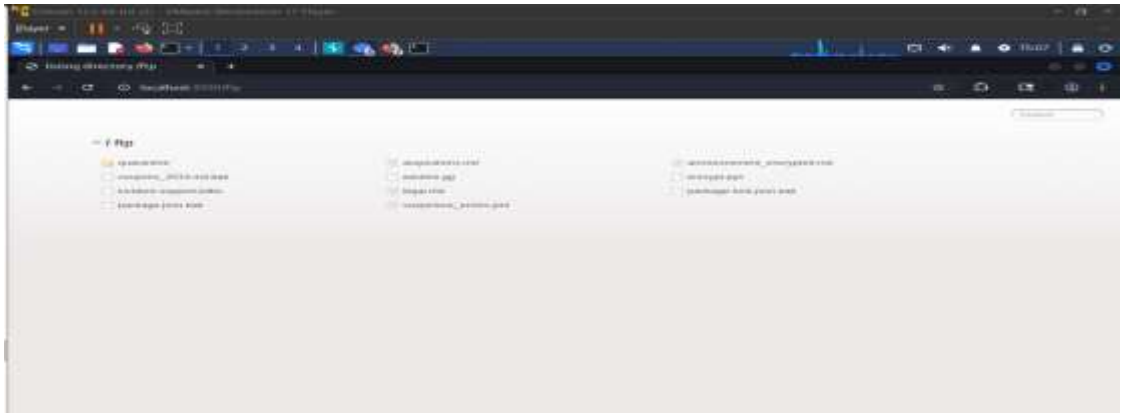
Description:
The application exposes sensitive files through an unprotected /ftp directory, allowing unauthorized users to access confidential data.

Impact (Medium Risk):
Sensitive files such as backups and configuration data may be accessed by attackers, potentially leading to further exploitation.

Steps to Reproduce:

1. Open browser and navigate to: http://localhost:3000/ftp
2. Browse the available files
3. Access downloadable files without authentication

Screenshot Explanation:



04 - Broken Authentication (No Rate Limiting)

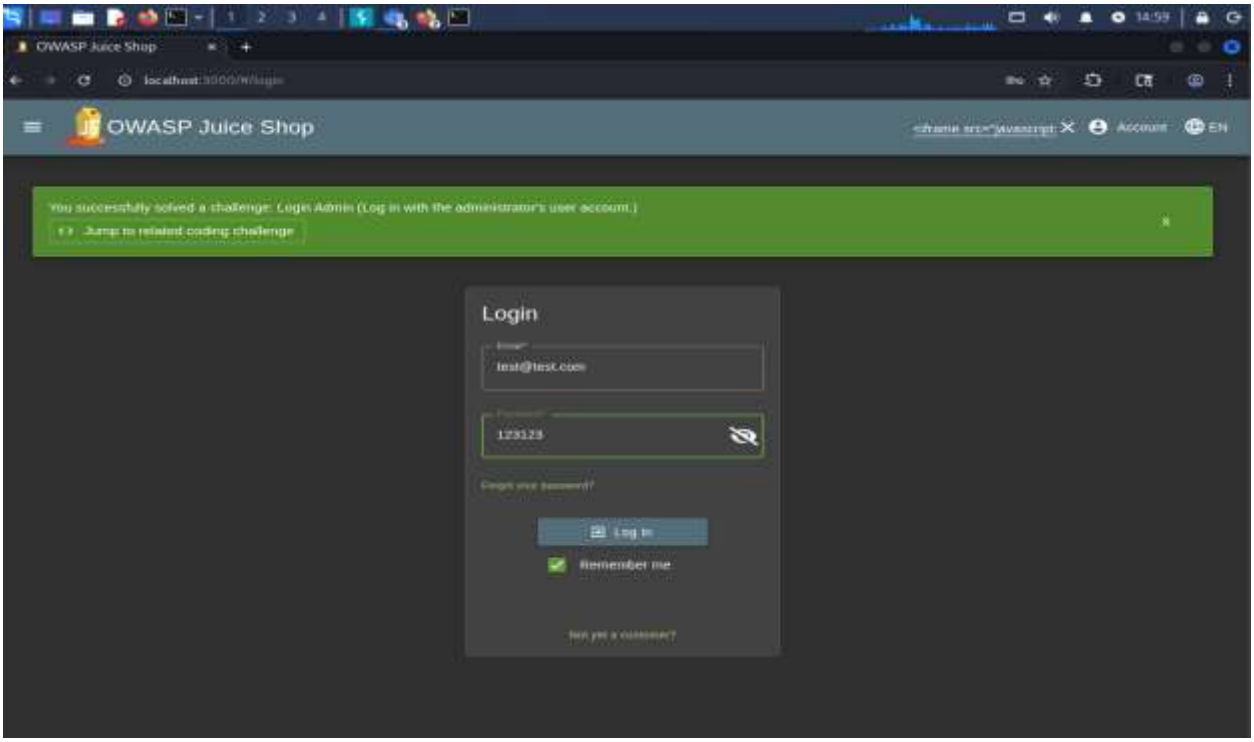
Description:
The application does not implement rate limiting or account lockout mechanisms, allowing unlimited login attempts.

Impact (Medium Risk):
Attackers can perform brute-force attacks to guess user credentials.

Steps to Reproduce:

1. Navigate to login page
2. Attempt multiple login requests with different passwords
3. Observe that no restrictions or delays are applied

Screenshot Explanation:



05 - Security Misconfiguration

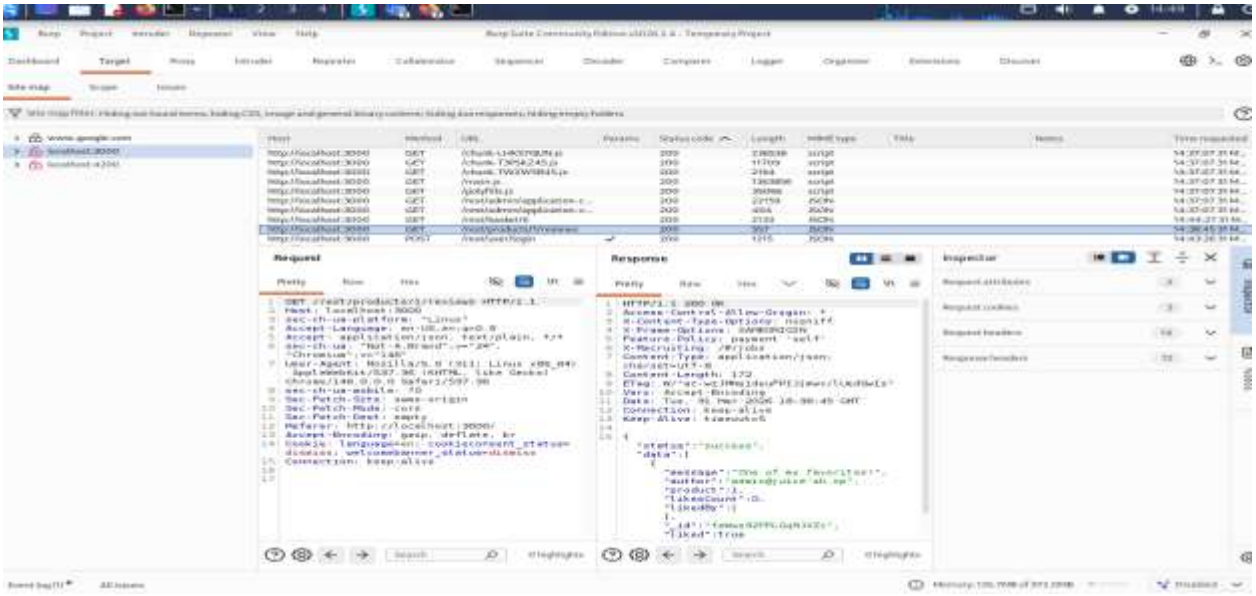
Description:
The application reveals excessive information through HTTP responses and error messages, indicating improper security configuration.

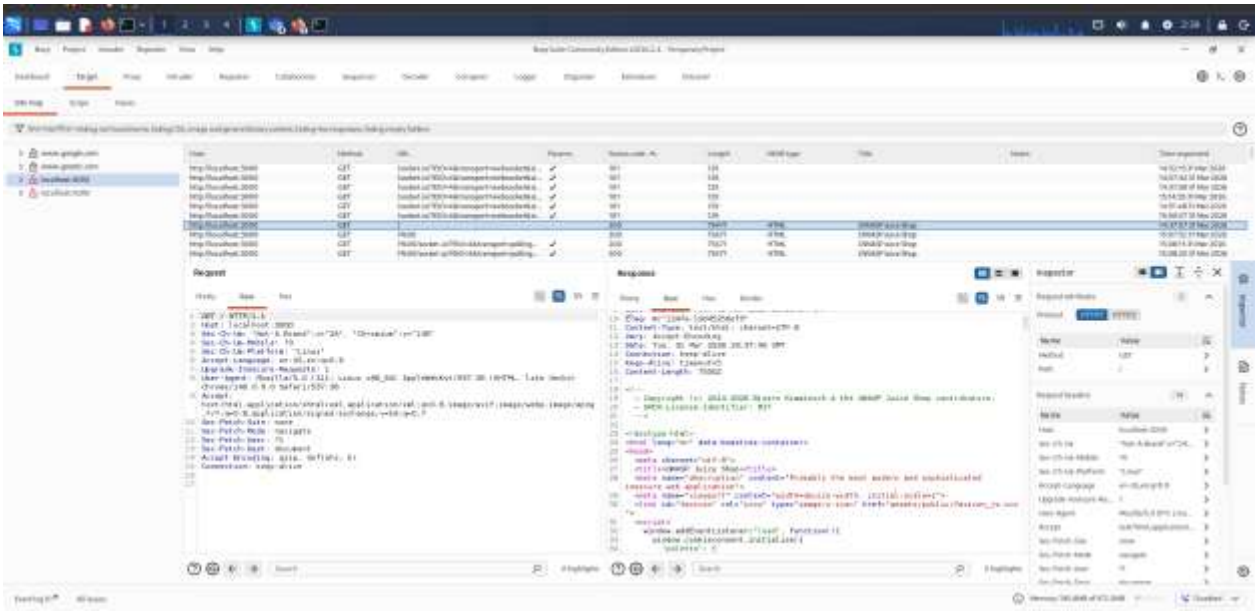
Impact (Low Risk):
Attackers can use this information to understand system architecture and craft targeted attacks.

Steps to Reproduce:

1. Intercept a request using Burp Suite
2. Analyze the HTTP response
3. Observe exposed headers and internal details

Screenshot Explanation:





06 - Broken Access Control

Description:
The application fails to properly enforce access control restrictions, allowing unauthorized users to access administrative endpoints.

Impact (High Risk):
Attackers can access restricted data and perform privileged actions without authorization.

Steps to Reproduce:

1. Intercept a request in Burp Suite
2. Modify header or request parameter:

X-User-Email: ' OR 1=1--

3. Forward the request
4. Observe access to admin-level data

Screenshot Explanation: